
NVIDIA
S3oRDMA and cuObject

S3



Amazon S3

- The Amazon Simple Storage Service (S3) is a popular object storage technology.
- Its scalability, availability, security and simplicity made a popular for data storage.
- The S3 interface became a standard and is now used by different object storage implementations.

Problem statement

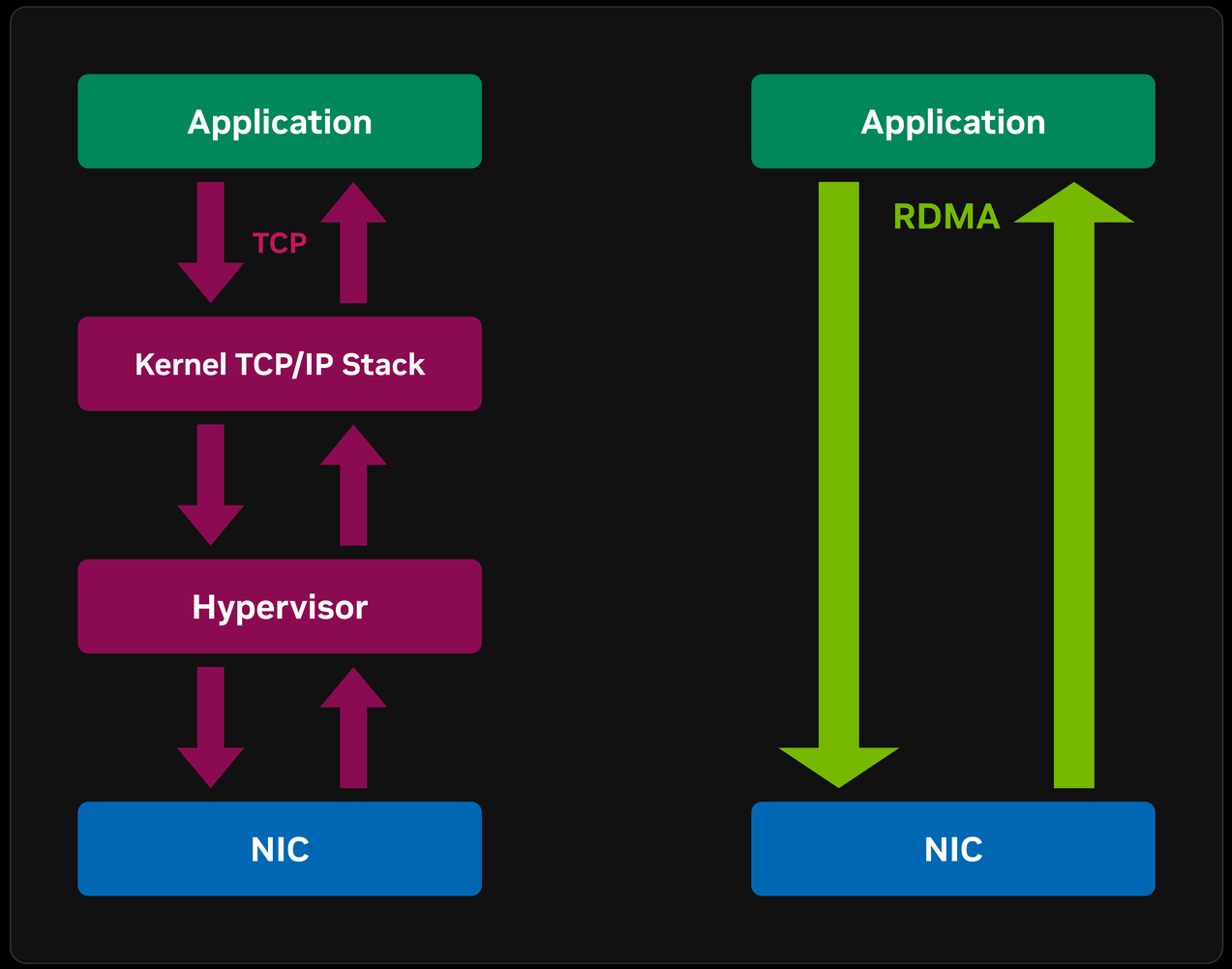
- S3 is HTTP based, which makes it unsuitable for worloads requiring low latency.
- AI inference is such a workload.

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RDMA - Remote Direct Memory Access

RDMA

- Kernel Bypass
- CPU offload
- Zero-copy



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cuObject - S3oRDMA for CUDA

The **cuObject** library enables direct data transfers between GPU memory and S3 compatible object storage using RDMA (= GPUDirect Storage for Objects).

Storage partners object server integration offers RDMA data source, sink and handles buffer registration.

